

Here's a comparison of best- and worst-case time complexities for the most common sorting algorithms:

Algorithm	Best-Case Time	Worst-Case Time
Bubble Sort	$O(n)$	$O(n^2)$
Insertion Sort	$O(n)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n^2)$
Heap Sort	$O(n \log n)$	$O(n \log n)$
Shell Sort	$O(n \log n)$	$O(n^2)$
Counting Sort	$O(n + k)$	$O(n + k)$
Radix Sort	$O(n \times k)$	$O(n \times k)$
Bucket Sort	$O(n + k)$	$O(n^2)$